electric fields

Important thing to note:
Electric fields are derived from **arrangements of electric charge**; magnetic fields come from arrangements of magnetic poles. Magnetic poles are always in pairs. Single magnetic monopoles have never been observed.

In general, field strength is defined as:

$$\text{field strength} = \frac{\text{force acting on an object in a field}}{\text{magnitude of the property that responds to the field}}$$

This definition is universal to **electric** and **gravitational** fields.

Hence, **electric field strength** E is:

$$E = \frac{F}{q}$$

where F is the force acting on a small test charge of charge magnitude q.

$$\therefore \text{The force } F \text{ acting on a charge } q \text{ is } F = qE$$

E is a **vector**. It acts in the direction of the force that would act on a **positive point test charge**. The unit of E is $\mathrm{N\ C^{-1}} \equiv \mathrm{kg\ ms^{-3}A^{-1}}$.

E can also be expressed as $E = \frac{V}{d}$, where V is the potential difference between two points and d is the distance between them. Hence, the unit of E is also $\mathrm{V\ m^{-1}} \equiv \mathrm{N\ C^{-1}}$.

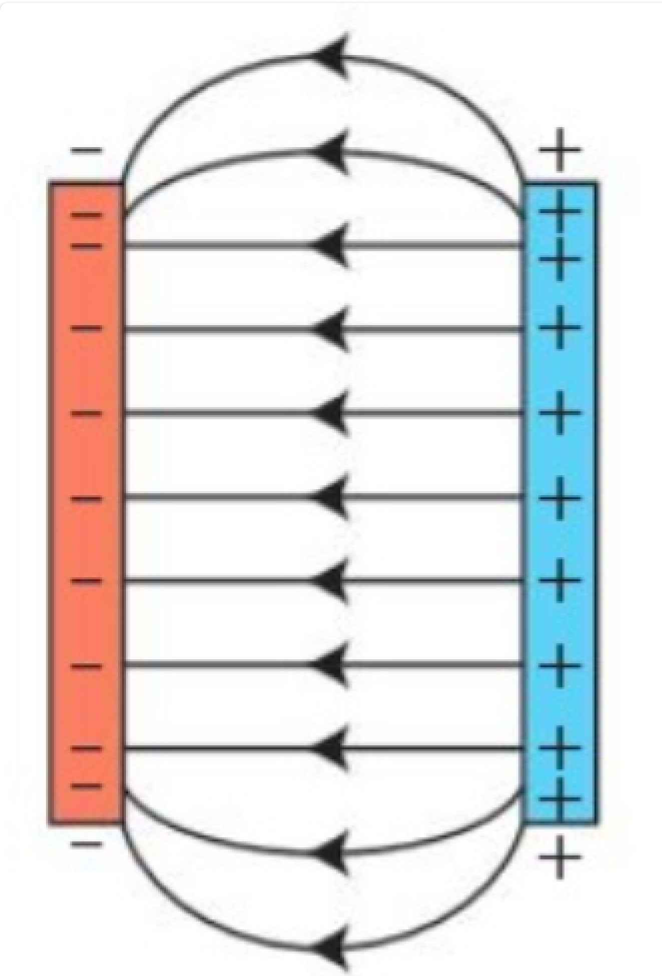
As seen in [gravitational fields](), E is also the [**electric potential**]() **gradient**:

$$E = -\frac{\Delta V_\mathrm{e}}{\Delta r}$$

Electric field lines can be used to show the shape of electric fields.
Arrows on the lines are required to show the direction in which a **positive charge** would move when free to do so.

Conventions for drawing electric field lines:

- Start and finish only on charges of opposite sign
- An arrow shows the field direction
- Lines close together $\implies$ field is stronger **or** the electric field line density is large
- Lines further apart $\implies$ field is weaker **or** the electric field line density is small
- Lines never cross
- Lines meet conducting surfaces at 90°



In the image above, $\exists$ an electric field between parallel (and identical) charged plates that are directly opposite each other. The field lines are **equally spaced** to signify that the field between the plates is **uniform**. However, this changes at the edge of the plates as shown by the gradual increase in line separation as the **field weakens**. This weakening is known as an **edge effect**.

This kind of electric field was used in [Millikan's oil drop experiment].